

4.Provision an AWS EC2 Instance Using Terraform

Aim

To install and configure Terraform and use Terraform Infrastructure as Code (IaC) to provision an AWS EC2 virtual machine.

Prerequisite:

Understand the basic concept

- **AWS** → Cloud platform that provides computing resources.
- **EC2** → A virtual computer/server provided by AWS.
- **Terraform** → Tool used to create and manage infrastructure using configuration files.
- **IaC** → Defining infrastructure using code instead of manually creating it.

The overall process is:

Terraform Code → Terraform → AWS → EC2 Instance

Create an AWS account

1. Open the [AWS website](#).
2. Create/sign in to an AWS account.
3. Open the **AWS Management Console**.
4. Select a region, for example **Asia Pacific (Mumbai) – ap-south-1**.

Procedure:

Install the AWS Command Line Interface and verify:

```
aws --version
```

Configure AWS:

```
aws configure
```

Enter:

AWS Access Key ID:

AWS Secret Access Key:

Default region name: ap-south-1

Default output format: json

Verify the configuration:

```
aws sts get-caller-identity
```

Install Terraform

Download and install Terraform from [HashiCorp Terraform](#).

Verify:

```
terraform --version
```

Create a Terraform project

Create a folder:

```
mkdir terraform-aws-ec2
cd terraform-aws-ec2
```

Create a file called:

Main.tf

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}

provider "aws" {
  region = "ap-south-1"
}

resource "aws_instance" "my_server" {
  ami           = "ami-0f58b397bc5c1f2e8"
  instance_type = "t3.micro"
  tags = {
    Name = "Terraform-EC2"
  }
}
```

Initialize Terraform

Run:

```
terraform init
```

Validate the configuration

```
terraform validate
```

Expected result:

Success! The configuration is valid.

Preview what Terraform will create

```
terraform plan
```

Create the EC2 instance

Run:

```
terraform apply
```

Terraform will ask for confirmation:

Enter a value: yes

Verify the instance

Go to:

AWS Console → EC2 → Instances

You can see:

Terraform-EC2

Result

Thus, an AWS EC2 virtual machine was successfully created, modified, verified, and destroyed using Terraform.